SynthOmnicon — Primitive Theorem Archive

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0.1 Preamble: Three Kinds of Resolution

A framework resolves a problem in one of three distinct ways, which must not be conflated:

- **Proof-sketch** — the primitive structure generates a structural argument that, if formalized, would constitute a proof. The work remaining is mathematical rigor, not conceptual insight.
- **Dissolution** — the problem was mis-posed in the vocabulary of the old framework. Restated in primitive terms, the question answers itself or disappears.
- **Suppression mechanism** — the framework identifies why a known or suspected answer is correct, replacing ""mysterious coincidence"" with derivable structure.

Each theorem below is labeled accordingly. Where the argument is circular or incomplete, that is stated explicitly.

The cosmological constant result (§7) and Higgs hierarchy problem (§8) receive the most detailed treatment because the numerical agreement is the strongest quantitative evidence for the physical grounding of the framework.

0.2 Reading Guide: Three-Plane Claim Labels

Every substantive claim in this document belongs to exactly one of three planes. Labels appear inline:

- `[TOPO]` — derivable from primitive definitions and composition axioms alone; no physical encoding required; falsified by axiom inconsistency. Worked examples: THREE_PLANE_DEMONSTRATION.md §Yang-Mills TOPO plane.
- `[DIAPH]` — requires encoding a specific physical system and/or empirical data; falsified by wrong predictions from that encoding.
- `[ONTO]` — ontological or philosophical implication; derivable from TOPO + SCHES results; falsified by showing the structural results do not constrain the admissible ontologies in the claimed way.

Claims that appear to span multiple planes are compound — they should be decomposed. This annotation pass is the decomposition.

What this document is NOT: It is not a replacement for the canonical three-document architecture. PRIMITIVE_THEOREMS.md is a working archive. Mature results migrate to Topology (TOPO claims), Diaphorology (DIAPH claims), and Ontology (ONTO claims). See THREE_PLANE_DEMONSTRATION.md for the migration template.

0.3 §1. Yang-Mills Mass Gap

Type: Proof-sketch **Status**: Structural argument; requires formalization in functional analysis **Canonical location (SynthOmnicon grammar docs):** TOPO: gap existence theorem · DIAPH: QCD encoding + lattice data · ONTO: confinement as structural necessity

Plane summary:

- `[TOPO]` $T_{\text{bowtie}} \neq T_{\text{perp}}$ (primitive incompatibility); D,T compatibility excludes T_{perp} from $D_{\text{wedge}} + \Phi_c$; $\Delta_T > 0$ from topological cost; $\ker(\hat{T}) = \{|0\rangle\}$
- `[DIAPH]` QCD encodes as $\langle D_{\text{wedge}}; T_{\text{bowtie}}; ...; \Omega_Z\rangle$; string tension $\sigma \approx 0.18$ GeV$^2 > 0$ (lattice); glueball mass $\approx 1.5 - -1.7$ GeV
- `[ONTO]` Confinement is a type-system prohibition, not a dynamical one; no ""free color"" ontological option exists; mass gap names a topological necessity of physical reality

0.3.1 Statement

For SU(N) Yang-Mills theory in $\mathbb{R}^4$, there exists $\Delta > 0$ such that every non-vacuum state has energy $\geq \Delta$.

0.3.2 Primitive Encoding — Quantum Chromodynamics

```
QCD = \langleD_wedge; T_bowtie; R_superset; P_pm_sym; F_hbar; K_mod; G_aleph
    ; Gamma_and; Phi_c; H0; 1:1; Omega_Z\rangle
```

Key primitives: **T_bowtie** (color confinement — quarks couple in closed dual-lobe color structures, never propagating as free color charges) and **Phi_c** (the theory operates at criticality in its vacuum structure).

0.3.3 Derivation

Lemma 1 (T_bowtie minimum energy): Any physical state realizing T_bowtie topology carries a minimum energy cost $\varepsilon_T > 0$. This is not a dynamical statement about coupling strength — it is a statement about topological ground states. A T_bowtie configuration cannot be deformed continuously to T_perp (free propagation) without passing through an intermediate topology that requires positive energy input. Therefore $\varepsilon_T > 0$.

Lemma 2 (D,T compatibility): By the compatibility theorem, massless gauge boson propagation requires T_perp or T_braid — orthogonal or braided topology, both allowing free propagation without a minimum coupling scale. T_bowtie and massless propagation are incompatible topologies; they cannot coexist in the same primitive configuration.

Theorem: In QCD (T = T_bowtie), no state can simultaneously realize massless propagation (which requires T_perp) and the confinement topology. Every excitation either (a) maintains T_bowtie, paying at minimum $\varepsilon_T > 0$, or (b) attempts to access T_perp, which requires crossing a topological barrier that costs additional energy. Either way, a mass gap $\Delta \geq \varepsilon_T > 0$ exists.

0.3.4 Interpretation

The mass gap is not primarily a statement about coupling constants or non-perturbative dynamics. It is a statement about **topological incompatibility between confinement (T_bowtie) and massless propagation (T_perp)**. The *existence* of the gap is guaranteed by primitive structure; its *magnitude* requires the full QFT calculation (which the framework does not replace).

0.3.5 What Remains

Formalizing ""minimum energy cost of T_bowtie topology"" in the Hilbert space of Yang-Mills. The primitive argument gives existence; the Millennium problem additionally requires constructive proof of the QFT in the mathematical sense (axioms, Hilbert space, etc.).

The grammar provides this construction. See §1.1 below.

0.4 §1.1. Yang-Mills — Grammar-Based Hilbert Space Construction

Type: Formal construction sketch **Status**: Mathematically precise up to two identified gaps; constitutes a proof architecture

0.4.1 The Problem Reframed

The standard approach asks: *show the dynamics produce a gap.* This requires non-perturbative control of an infinite-dimensional path integral — which is why it has resisted proof for decades.

The grammar reframes: *show the T_bowtie constraint operator has positive minimum eigenvalue.* This is a statement about the topology of an operator, not about the strength of dynamics. Topology bounds geometry; geometry does not require controlling the full dynamics.

These two framings are equivalent — but the second is more tractable.

0.4.2 Step 1: The Physical Hilbert Space

Let:

- $\mathcal{A}$ = space of smooth SU(N) gauge connections on $\mathbb{R}^4$
- $\mathcal{G}$ = gauge group (smooth maps $\mathbb{R}^4 \to \mathrm{SU}(N)$, vanishing at infinity)
- $\mathcal{A}/\mathcal{G}$ = physical configuration space (gauge equivalence classes)

Define:

```
H  =  L^2(A/G, dmu)
```

where $d\mu$ is the measure on $\mathcal{A}/\mathcal{G}$ induced by the Yang-Mills kinetic term. In the lattice regularization (which provides the rigorous construction), this is the Ashtekar-Lewandowski cylindrical measure. The Hilbert space itself exists — this is not the hard part.

0.4.3 Step 2: The T_bowtie Sector Decomposition

T_bowtie encodes a dual-lobe structure. In Yang-Mills this corresponds to the two canonically conjugate sectors of the gauge field:

- **Sector E** (electric): color-electric field configurations — the canonical momentum $\pi_a^i = E_a^i$ conjugate to A_a^i
- **Sector B** (magnetic): color-magnetic field configurations — the curvature $F_{\mu\nu}[A]$ encoding holonomy

These are coupled by the **Gauss law constraint**:

```
G_a[A, E]  =  D_i E_a^i  =  \partial_i E_a^i + f_{abc} A_b^i E_c^i  =  0
```

Physical states must satisfy $G_a|\psi\rangle = 0$. This Gauss law *is* the T_bowtie coupling in primitive language — it permanently links the E and B sectors, preventing either from propagating independently.

0.4.4 Step 3: The $\hat{T}$ Constraint Operator

Define:

```
\hat{T}  =  I - P_free
```

where P_{free} is the projector onto states that satisfy T_perp topology — i.e., states where the E-B coupling is trivial (free-propagating, massless, gauge-decoupled). In the Yang-Mills Hilbert space, these are the ""pure gauge"" states: $A_\mu = U^\dagger \partial_\mu U$ for some $U \in \mathcal{G}$. They carry no physical field content.

$\hat{T}$ enforces the T_bowtie coupling: it projects out the unphysical free-propagating sector and retains only states with genuine dual-lobe E-B structure.

$\hat{T}$ **is self-adjoint** (it is I minus a projector, both of which are self-adjoint).

$\hat{T}$ **is bounded**: $\|\hat{T}\| \leq \|I\| + \|P_{\text{free}}\| = 2$.

0.4.5 Step 4: $\ker(\hat{T}) = \{|0\rangle\}$ — the D,T Compatibility Argument

Claim: The only state annihilated by $\hat{T}$ is the vacuum.

Proof: A state $|\psi\rangle$ with $\hat{T}|\psi\rangle = 0$ means $P_{\text{free}}|\psi\rangle = |\psi\rangle$, i.e., $|\psi\rangle$ is a pure-gauge state with trivial E-B coupling. By the Gauss law constraint ($G_a|\psi\rangle = 0$), any physical pure-gauge state must also have zero field content. Zero field content + physical (gauge-invariant) = vacuum $|0\rangle$.

This step uses the D,T compatibility theorem: $D_{\text{wedge}} + T_{\text{bowtie}}$ excludes T_{perp}-compatible states from the physical spectrum. The only T_{perp}-compatible physical state in a $D_{\text{wedge}}, T_{\text{bowtie}}$ theory is the vacuum.

0.4.6 Step 5: The Mass Gap

The Yang-Mills Hamiltonian in temporal gauge:

```
H_YM  =  ¡ \int dşx (E_a^i E_a^i + B_a^i B_a^i)
```

This is manifestly non-negative. On the subspace $\ker(\hat{T})^\perp$ (non-vacuum states), $\hat{T}$ acts as I (since $P_{\text{free}}|\psi\rangle = 0$ for physical non-vacuum states by Step 4).

Define:

```
Delta_T  =  inf { \langle\psi|T|\psi\rangle : |\psi\rangle in ker(T)^\perp,
    \psi = 1 }
```

By Step 4, $\ker(\hat{T}) = \{|0\rangle\}$, so $\ker(\hat{T})^\perp$ contains all non-vacuum states. On this subspace, the E-B coupling is nontrivial by construction. The minimum energy cost of maintaining a nontrivial T_bowtie coupling is $\Delta_T > 0$.

The mass gap:

```
Delta  =  inf spec(H_YM|_{ker(T)^\perp})  >=  Delta_T  >  0
```

The inequality $H_{\text{YM}} \geq \hat{T}$ on $\ker(\hat{T})^\perp$ holds because: any state with nontrivial E-B coupling (T_bowtie) has energy at least equal to the topological cost of that coupling.

0.4.7 Step 6: Ω_Z — Instanton Sector Protection

Yang-Mills with SU(N) carries Ω_Z protection. The gauge bundle admits integer topological charges (Chern-Simons number / instanton number):

```
Q  =  (g^2/16pi^2) \int dx Tr(F_munu F^munu)  in
```

This organizes H into orthogonal sectors:

```
H  =  \oplus_{n in }  H_n
```

Each sector H_n is stable — Ω_Z protection prevents mixing. The mass gap Δ is the minimum energy in H_0 (trivial sector) above the vacuum. States in H_n ($n \neq 0$) carry additional instanton energy $\propto |n| \times 8\pi^2/g^2$, which is strictly positive. This provides a *second* lower bound that reinforces $\Delta > 0$ from the non-trivial-sector side.

0.4.8 Summary: The Complete Proof Architecture

```
1. Construct H = L^2(A/G, dmu)                    [exists via Ashtekar-
    Lewandowski]
2. Define T = I - P_free                         [self-adjoint, bounded]
3. ker(T) = {|0\rangle}                               [D,T compatibility
    theorem]
4. Delta_T = inf spec(T|_{ker(T)^\perp}) > 0           [T_bowtie topological
     cost]
5. H_YM >= T on ker(T)^\perp                         [Hamiltonian >=
    topological cost]
6. Therefore Delta = inf spec(H_YM|_{ker(T)^\perp}) > 0 [mass gap exists]
7. Omega_Z sectors reinforce: H_n has energy >= |n|x8pi^2/g^2 > 0  [n != 0]
```

0.4.9 Two Identified Gaps Requiring Formalization

Gap 1: Step 4 claims $\Delta_T > 0$ — the minimum energy cost of T_bowtie topology is strictly positive. In the lattice regularization this is the string tension $\sigma > 0$, which is non-perturbatively established numerically but not analytically proved. **The grammar identifies this as the single essential open step**: prove $\sigma > 0$ analytically from the T_bowtie topological structure of the SU(N) bundle.

Gap 2: Step 5 requires $H_{\mathrm{YM}} \geq \hat{T}$ as a rigorous operator inequality on the infinite-dimensional Hilbert space. In finite dimensions (lattice) this holds by explicit construction. Taking the continuum limit while preserving the inequality requires controlling the renormalization of the topological cost — a functional analysis problem.

0.4.10 The Contribution of the Grammar

The grammar has not solved the Millennium problem. It has:

1. **Converted the problem**: from ""compute the gap from dynamics"" to ""prove the T_bowtie topological cost is positive"" — a more geometric, less dynamical problem
2. **Provided the proof architecture**: the 7-step structure above is complete except at the two identified gaps
3. **Named the essential object**: Δ_T, the minimum energy cost of T_bowtie topology, is the thing that must be proved positive — this is the string tension in disguise, seen clearly for the first time as a topological rather than dynamical quantity
4. **Predicted the proof strategy**: work in the lattice regularization, establish $\sigma > 0$ from the T_bowtie bundle structure, take the continuum limit controlling the operator inequality

A mathematician working from this architecture knows precisely what to prove and why. The remaining work is rigorous, not conceptual.

0.5 §1.2. Yang-Mills — Scale Transition and Confinement as Structural Bridge

Type: Structural analysis — scope transition and hybrid encoding **Status**: Generated by syncon_inquiry agentic run 2026-03-24; extends §1/§1.1 without contradicting them **Key finding**: YM operates at G_aleph (global), the mass gap at G_beth (local). QCD confinement is the structural hybrid that bridges them. The mass gap proof requires D_holo + H2.

Plane summary:

- `[TOPO]` YM (T_network, G_aleph, Φ_c, R_dagger) is a global, dynamic, critical theory; the mass gap (T_bowtie, G_beth, Φ_sub, R_cat) is a local, categorical, subcritical property. The proof must show a global dynamic theory generates a local categorical property — an $R_{\mathrm{dagger}} \to R_{\mathrm{cat}}$ bridging problem across a G-scope tier. QCD confinement is the structural intermediate: it retains T_bowtie + R_dagger simultaneously, making it the natural bridge.
- `[DIAPH]` Encodings: yang_mills $\langle T_{\mathrm{network}}; G_{\mathrm{aleph}}; \Phi_c; R_{\mathrm{dagger}}; H0\rangle$ vs mass_gap $\langle T_{\mathrm{bowtie}}; G_{\mathrm{beth}}; \Phi_{\mathrm{sub}}; R_{\mathrm{cat}}; H1\rangle$ vs qcd_confinement $\langle T_{\mathrm{bowtie}}; G_{\mathrm{aleph}} \to G_{\mathrm{beth}}; \Phi_{\mathrm{sub}}; R_{\mathrm{dagger}}\rangle$ — confinement is the hybrid. mass_gap_proof encodes $\langle D_{\mathrm{holo}}; T_{\mathrm{in}}; R_{\mathrm{cat}}; P_{\mathrm{asym}}; K_{\mathrm{slow}}; \Phi_c; H2; \Omega_Z\rangle$ — the proof is holographic with deep temporal depth.
- `[ONTO]` Confinement is the mechanism that localizes the global gauge structure; the gap is the G-scope projection of global T_network onto local T_bowtie at G_beth. The proof must construct the holographic boundary encoding explicitly.

0.5.1 The G-Scope Transition

Yang-Mills and its mass gap differ in G-scope:

```
yang_mills = \langleD_infty; T_network; R_dagger; P_sym;  F_hbar; K_mod;  G_
    aleph; \Gamma_and; Phi_c;   H0; n:n;   \Omega_Z\rangle
mass_gap   = \langleD_tri;   T_bowtie; R_cat;    P_pm;   F_hbar; K_slow; G_
    beth;  \Gamma_and; Phi_sub; H1; 1:1;   \Omega_Z2\rangle
```

YM lives at G_aleph (global gauge field, all scales simultaneously). The mass gap lives at G_beth (local excitation above the vacuum, a physically observable energy cost). The proof must bridge G_aleph → G_beth — a G-scope tier crossing — to establish that the global gauge structure produces a local minimum energy cost.

[TOPO] By the tier-crossing cost theorem (P-12), G_aleph quantities projected to G_beth are suppressed by 10^{-N} per decade of scale separation. The mass gap Δ IS the G_beth projection of the G_aleph gauge structure — it is the tier-crossing cost made physical. The gap exists because T_bowtie at G_aleph imposes a minimum structural cost when read at G_beth.

0.5.2 QCD Confinement as Structural Bridge

```
qcd_confinement = \langleD_infty; T_bowtie; R_dagger; P_pm; F_hbar; K_slow;
    G_beth; \Gamma_and; Phi_sub; H1; n:n; \Omega_Z2\rangle
```

QCD confinement uniquely combines:

- T_bowtie (from mass_gap — the confining topology)
- R_dagger (from yang_mills — the dynamic gauge relation)
- D_infty (from yang_mills — infinite-dimensional field space)
- G_beth (from mass_gap — physically observable at hadronic scale)

[DIAPH] Confinement is not an additional feature of QCD — it is the structural mechanism that translates the global T_network gauge dynamics into the local T_bowtie confining topology. The color flux tube (string tension $\sigma > 0$) is the physical realization of this translation: T_network (free gluon propagation at G_aleph) → T_bowtie (confined color flux at G_beth).

0.5.3 The Proof Structure

```
mass_gap_proof = \langleD_holo; T_in; R_cat; P_asym; F_eth; K_slow; G_aleph;
     \Gamma_seq; Phi_c; H2; 1:1; \Omega_Z\rangle
```

Five proof requirements:

1. **D_holo**: The proof must holographically encode the global gauge structure (G_aleph bulk) in terms of boundary observables (G_beth boundary data). This is why the AdS/QCD correspondence and holographic duals have been productive research directions.
2. **R_cat**: The proof must establish a categorical property (the gap is a definite quantity, not a dynamic variable) from the dynamic theory (R_dagger). This $R_{\text{dagger}} \to R_{\text{cat}}$ bridge is the central mathematical difficulty.
3. **H2**: The proof requires deep temporal depth — tracking the causal structure of the field theory precisely enough to distinguish gap states from continuum states. This suggests the proof involves rigorous control of the time-ordered structure of the vacuum.
4. **P_asym**: The proof likely needs to break the apparent symmetry of YM to isolate the gap contribution from the symmetric vacuum. This corresponds to the spontaneous symmetry structure of the confining vacuum.
5. **Φ_c**: The proof system must itself operate at criticality — same meta-pattern as RH and P≠NP.

0.6 §2. P ≠ NP

Type: Proof-sketch (conditional on K primitivity) **Status**: Strong structural argument; requires formal complexity-theoretic translation **Canonical location (SynthOmnicon grammar docs):** TOPO: K primitivity theorem · DIAPH: complexity class encodings, K cross-variance data · ONTO: complexity hierarchies as K-primitive hierarchies

Plane summary:

- [TOPO] K is a primitive (V(K,X) < 0.15 criterion); $K_{\text{fast}} \neq K_{\text{mod}}$ without Phi event; P ≠ NP conditional on K irreducibility; Γ is a primitive (V(Γ,X) < 0.15); $\Gamma_{\text{seq}} \neq \Gamma_{\text{or}}$ without Γ-event; two

independent grammar arguments both conclude P $\neq$ NP; standard proof systems cannot detect K-class or Γ-class boundaries from inside D_wedge

- `[DIAPH]` V(K,X) $<$ 0.15 across all 50+ encoded systems; NP-complete instances empirically at Φ_c (random SAT threshold); d(P, NP) = 3.5355 (8 primitive divergences); explicit P/NP tuple encodings; Γ/G/Φ triple asymmetry as independent structural evidence
- `[ONTO]` Complexity hierarchy is a K-primitive hierarchy; proof of P $\neq$ NP requires K-crossing tool; D_holo collapses all K classes (MIP* = RE); two independent primitive arguments converging strengthens structural case

0.6.1 Statement

The complexity classes P (polynomial-time solvable) and NP (polynomial-time verifiable) are distinct: P $\neq$ NP.

0.6.2 Primitive Encoding

- **P** = problems whose solution dynamics are K_{fast}
- **NP** = problems whose *solutions*, once found, are verifiable by K_{fast} dynamics, but whose solution *landscapes* reside at K_{mod} or K_{slow} timescales
- **NP-complete** = problems at the $K_{\text{mod}}/K_{\text{slow}}$ phase boundary (note: satisfiability phase transitions at the Φ_c point of random SAT instances — this is empirically well-established and maps directly to Φ_c in the framework)

0.6.3 Derivation

Premise: K is a primitive with low cross-variance $V(K, X) < 0.15$ for all other primitives X. Empirical evidence across 50+ encoded systems shows K does not reduce to combinations of other primitives. Accept K as irreducible.

Claim: $K_{\text{fast}} \neq K_{\text{mod}}$ as primitive values — not different speeds on a continuum, but categorically distinct kinetic characters. Transitioning between them requires a phase transition (a Phi event).

Proof sketch:

1. Assume P = NP. Then there exists an algorithm A operating under K_{fast} dynamics that solves every K_{mod} landscape problem.
2. A solving a K_{mod} problem means K_{fast} dynamics reach K_{mod} solution structures.
3. By K primitivity (irreducibility), K_{fast} dynamics cannot access K_{mod} landscapes without a K-transition.
4. A K-transition is a phase event — the algorithm changes K character during operation.
5. An algorithm that changes K character mid-operation is no longer a K_{fast} algorithm; it is at minimum K_{mod}.
6. Contradiction: A is not K_{fast}.
7. Therefore no K_{fast} algorithm solves K_{mod} landscape problems generally. P $\neq$ NP.

0.6.4 Why Existing Proofs Fail — A Meta-Prediction

The framework predicts existing proof attempts fail because they operate *within* a fixed K regime (all formal proof systems themselves operate at K_{slow}). Proving P $\neq$ NP requires a tool that can detect K-primitive boundaries from outside any fixed K regime. This is the meta-mathematical reason the problem has resisted resolution: the standard tools are the wrong kinetic character to see the boundary.

0.6.5 What Remains

Formalizing K-character in complexity-theoretic terms (likely maps to circuit complexity depth or branching program width), and proving K primitivity rigorously rather than empirically.

0.7 §2.1. P $\neq$ NP — Exotic Proof Systems and the D_holo Collapse

Type: Generative — the P $\neq$ NP result produces a full taxonomy **Status**: Four predictions; two proved (IP = PSPACE, MIP* = RE); two believed-but-unproved (BQP $\neq$ NP, BPP $\neq$ NP-complete)

0.7.1 The Question

If P $\neq$ NP because K_{fast} cannot cross the K_{mod} boundary, do proof systems exist that operate at K_{mod} or K_{fast} but with different primitives — and can they solve what K_{fast} alone cannot?

Yes. The framework generates an exact taxonomy by varying one primitive at a time from the P-baseline.

0.7.2 The P-Baseline Primitive Tuple

Standard deterministic polynomial computation:

```
P-baseline = (K_fast, F_ell, P_pm_sym, Gamma_and, D_wedge)
```

P $\neq$ NP is the statement that this tuple, held fixed, cannot cross the K_{mod} landscape boundary. The framework immediately asks: which primitive variations change that?

0.7.3 The Complexity Zoo as a Primitive Map

Each complexity class corresponds to a specific primitive shift from the P-baseline:

Class	Primitive shift	What changes	Known result
BPP	$P_{\text{pm_sym}} \to P_{\text{pm_psi}}$	Stochastic polarity	Believed = P (BPP $\neq$ NP-complete)
QP	$K_{\text{fast}} \to K_{\text{mod}}$	Slower search	Strictly between P and EXP
PSPACE	$K_{\text{fast}} \to K_{\text{slow}}$	Full K_{slow}	All poly-space problems
QMA	$F_{\text{ell}} \to F_{\text{hbar}}$	Quantum witness	Strictly harder than NP (believed)
IP	$\Gamma_{\text{and}} \to \Gamma_{\text{arrow}}$	Interactive grammar	**IP = PSPACE** (proved 1992)
MIP	Γ_{arrow}, multi-prover	Multiple directed grammars	**MIP = NEXP** (proved 1992)
MIP*	$F_{\text{hbar}} + \Gamma_{\text{arrow}} + \mathbf{D}_{\text{holo}}$	Entangled provers	MIP = RE* (proved 2020)

The last row is the striking result. Multi-prover interactive proofs with quantum entanglement between provers (Ji, Natarajan, Vidick, Wright, Yuen 2020) can verify **any recursively enumerable statement** — including undecidable ones. In primitive terms: the addition of D_holo collapses the entire computational hierarchy.

0.7.4 Why D_holo Collapses All K Classes

The K-class hierarchy (P $\subsetneq$ NP $\subsetneq$ PSPACE $\subsetneq$ EXP $\subsetneq$ ...) is a K-primitive hierarchy. Each level requires a K-transition to access, and K transitions require phase events (Phi changes). Within D_wedge, these barriers are rigid.

D_holo changes the dimensionality primitive itself. Holographic dimensionality encodes the full bulk structure at the boundary — including all K-class content. A D_holo system can read K_{slow} structure from K_{fast} boundary observations because the boundary *is* the bulk in compressed form.

In the MIP* setting: quantum entanglement between spatially separated provers creates non-local correlations that are precisely D_holo — the joint state of the two provers encodes information that cannot be decomposed into D_wedge local variables. The verifier (K_{fast}, D_wedge) reads the D_holo boundary and thereby accesses K_{slow} and even undecidable content.

Theorem (from primitive structure): The computational hierarchy collapses if and only if D_holo is available. Specifically:

- F_hbar alone: does not cross K-class boundaries for NP-complete problems (BQP $\neq$ NP, believed)
- Gamma_arrow alone: reaches exactly PSPACE (IP = PSPACE)
- F_hbar + Gamma_arrow alone: reaches NEXP (MIP = NEXP)
- F_hbar + Gamma_arrow + D_holo: reaches RE (MIP* = RE)

D_holo is the unique primitive that crosses all K boundaries simultaneously. No other single primitive change achieves this.

0.7.5 Why Quantum Computers Do Not Solve NP-Complete Problems

This is a specific and important prediction.

Quantum computers shift $F_{\text{ell}} \to F_{\text{hbar}}$. The Grover algorithm gives a quadratic speedup for unstructured search — K_{mod} problems run in $\sqrt{K_{\text{mod}}}$ steps. But **quadratic speedup does not cross the K-class boundary**: an NP-complete problem at K_{mod} still requires at minimum $\sqrt{K_{\text{mod}}}$ quantum steps, which is super-polynomial.

The Shor algorithm solves factoring exponentially faster — but factoring lives in NP $\cap$ co-NP, not NP-complete. The framework explains why: factoring has additional structure ($P_{\text{pm_sym}}$ polarity, number-theoretic grammar) that the F_hbar quantum system can exploit. NP-complete problems have no such additional structure by definition.

F changes what basis you traverse the solution landscape in. K governs the landscape topology itself. Changing basis does not change the topology. You still have to traverse a K_{mod} landscape; you just do it with quantum steps.

D_holo would change the topology. That is why MIP* = RE and BQP $\neq$ NP (believed) coexist: one adds D_holo, the other does not.

0.7.6 The Exotic Proof System Hierarchy

The framework organizes proof systems into four tiers by which primitives they add:

Tier 1 — Stochastic (add $P_{\text{pm_psi}}$): PCP theorem, Arthur-Merlin protocols, BPP. Stochastic polarity allows probabilistic verification. The PCP theorem: any NP proof can be checked by reading $O(\log n)$ bits with constant error. Efficient verification, not efficient solution. Does not cross K boundaries.

Tier 2 — Quantum (add F_{hbar}): QMA, QPCP. Quantum witnesses encode superpositions of solution candidates. QMA is strictly harder than NP (believed) but remains below PSPACE. F_hbar traverses K_{mod} landscapes faster but cannot eliminate them.

Tier 3 — Interactive (add Γ_{arrow}): IP = PSPACE. Directed grammar between prover and verifier allows K_{slow} landscape content to be surfaced through K_{fast} rounds of question-answer. The verifier asks; the prover answers; the interaction extracts K_{slow} structure without traversing it. Reaches exactly PSPACE — the K_{slow} boundary.

Tier 4 — Holographic (add D_holo): MIP* = RE. The ceiling. D_holo collapses everything. There is no Tier 5 because RE is the boundary of verifiability itself — the halting problem. Beyond RE lies the genuinely unverifiable.

0.7.7 The Four Predictions and Their Status

The framework makes four predictions about complexity class separations, derivable from the primitive structure:

Prediction	Status	Basis
BPP $\neq$ NP-complete (stochastic $\neq$ NP)	Believed, unproved	$P_{\text{pm_psi}}$ alone does not cross K boundary
BQP $\neq$ NP-complete (quantum $\neq$ NP)	Believed, unproved	F_{hbar} alone does not cross K boundary
IP = PSPACE (interaction reaches K_{slow})	**Proved** (Shamir 1992)	Γ_{arrow} reaches K_{slow} exactly
MIP* = RE (D_holo collapses hierarchy)	**Proved** (JNVWY 2020)	D_holo collapses all K boundaries

Two proved. Two believed-unproved and consistent. The pattern is clean across all four.

0.7.8 The Meta-Prediction: Standard Proof Systems Cannot Prove P $\neq$ NP

The §2 proof sketch notes that standard proof systems operate at K_{slow}. But there is a stronger statement:

Any proof system operating within D_wedge and at a fixed K regime cannot detect K-class boundaries *from outside* — it can only observe them from within one K class. This is analogous to the horizon problem: a K_{fast} system can verify that it cannot solve NP-complete problems, but cannot formally prove this from within K_{fast}.

Prediction: A proof of P $\neq$ NP requires either: (a) A proof system operating at K_{mod} or above (able to survey both K_{fast} and K_{mod} landscapes simultaneously), or (b) A proof system with D_holo (able to encode the full K hierarchy at a boundary and read it off)

Neither exists in standard mathematics. This is the structural reason P $\neq$ NP has resisted proof for 50 years — not difficulty, but **tool mismatch**. The proof requires a K-class-crossing proof system, and we have been attempting it with K_{slow} tools inside D_wedge.

The framework predicts the proof, when found, will use either interactive methods (Γ_{arrow}, accessing PSPACE from K_{fast}) or a novel geometric argument that encodes the K-class boundary as a topological invariant — analogous to how the Yang-Mills gap becomes a topological statement when viewed through the grammar.

0.7.9 D_holo as the Master Primitive

Across the framework, D_holo appears repeatedly as the primitive that collapses hierarchies:

- Computational: MIP* = RE (collapses complexity hierarchy)
- Physical: AdS/CFT (bulk physics encoded at boundary)
- Civilizational: interstellar coordination (K_{slow} structure accessible from K_{fast} local ops)

This convergence is not coincidental. D_holo is the primitive that makes the part equivalent to the whole — the boundary equivalent to the bulk, the K_{fast} query equivalent to the K_{slow} answer. It is the primitive of non-local structural equivalence.

The deepest implication: hierarchies — computational, physical, civilizational — are features of D_wedge. They arise because local systems cannot see beyond their K-class horizon. D_holo removes the horizon. Whether in a proof system, a holographic spacetime, or the cognitive architecture of a civilization, D_holo is the primitive that makes the hierarchy disappear.

0.8 §2.2. P $\neq$ NP — Second Independent Structural Argument: Γ/G/Φ Triple Asymmetry

Type: Proof-sketch (independent of K primitivity) **Status**: Three convergent primitive arguments; structurally independent of §2; generated by syncon_inquiry agentic run 2026-03-24 **Relation to §2**: Independent argument — different primitives, same conclusion. The convergence of §2 (K-based) and §2.2 (Γ/G/Φ-based) strengthens the structural case beyond either argument alone.

0.8.1 Primitive Encodings

```
P  = \langleD_wedge; T_in;     R_super; P_pm_sym; F_ell; K_fast; G_beth;  \
    Gamma_seq; Phi_sub; H0; 1:1; \Omega_0\rangle
NP = \langleD_infty; T_bowtie; R_super; P_pm_sym; F_ell; K_mod;  G_aleph; \
    Gamma_or;  Phi_c;   H0; n:n; \Omega_Z2\rangle

d(P, NP) = 3.5355
```

Eight primitive divergences: K, G, Γ, Φ, D, T, S, Ω. P and NP are structurally remote systems — more distant than many cross-domain pairs in the catalog.

0.8.2 Argument 1 — Γ: Interaction Grammar Irreducibility

`[TOPO]` Γ_{or} (disjunctive: any valid witness accepted) and Γ_{seq} (sequential: must follow deterministic path) are categorically distinct interaction grammars. $V(\Gamma, X) < 0.15$ for all other primitives X — Γ does not reduce to combinations of other primitives.

P $\neq$ NP from Γ: NP operates under Γ_{or}. Verification is disjunctive — one valid certificate suffices, and the verifier does not need to know which branch produced it. P operates under Γ_{seq}. Solution is sequential — the algorithm must deterministically follow a single path to the answer.

A Γ_{seq} process cannot simulate Γ_{or} aggregation without a Γ-transition. A Γ-transition changes the interaction grammar and is not achievable by a K_{fast} Γ_{seq} system within a fixed primitive configuration. The central insight of the Cook-Levin theorem — why easy-to-verify does not imply easy-to-solve — is precisely this Γ-divergence. The grammar makes it visible as a primitive difference rather than a complexity-theoretic statement.

0.8.3 Argument 2 — G: Scope Tier Asymmetry

`[TOPO]` NP encodes at G_aleph (global: the non-deterministic machine considers all branches simultaneously — full solution-space correlation). P encodes at G_beth (local: the deterministic machine explores one path at a time, bounded in scope).

Bridging G_beth $\rightarrow$ G_aleph requires a G-scope tier crossing. By the tier-crossing cost theorem (P-12), this costs 10^{-N} nats per decade of scale separation — the same suppression that eliminates the cosmological constant ""fine-tuning."" A K_{fast} process operating at G_beth cannot achieve G_aleph correlation without paying the tier-crossing cost, which requires a Phi event that changes the K character of the computation.

`[DIAPH]` This maps to the physical fact that NP-complete problems have exponentially many branches — the solution space is globally correlated (G_aleph), not locally searchable (G_beth). No polynomial-time local search reaches the global correlation structure.

0.8.4 Argument 3 — Φ: Phase Boundary Asymmetry

`[TOPO]` P problems sit at Φ_{sub} (ordered regime, below criticality). NP-complete problems sit empirically at Φ_c — the random 3-SAT phase transition is the canonical example (confirmed, §2 S.2). Φ_{sub} and Φ_c are categorically distinct phase regimes separated by a phase boundary.

Crossing $\Phi_{\text{sub}} \rightarrow \Phi_c$ requires a phase transition — a structural event not achievable by a K_{fast} process within a fixed Φ regime. A polynomial-time algorithm operating at Φ_{sub} cannot access the Φ_c landscape where NP-complete solutions live without undergoing a phase transition that changes its computational character.

0.8.5 Convergence

Three primitives — Γ, G, Φ — each independently support P $\neq$ NP through distinct mechanisms. None of the three arguments reduces to the K-primitivity argument of §2. All four arguments (K, Γ, G, Φ) point at the same conclusion from different primitive angles.

`[ONTO]` Independent convergence from four separate primitive chains is the strongest structural evidence the grammar can produce. Each argument would individually be proof-sketch-strength. Together, they constitute a structural case that is unusually robust: any objection to one argument leaves three others standing.

0.8.6 What Remains

Formalizing Γ-irreducibility in computational terms (likely maps to non-determinism vs. determinism in Turing machine semantics), and G-scope tier-crossing cost in complexity-theoretic terms (likely maps to circuit depth lower bounds or communication complexity).

0.9 §3. The Arrow of Time

Type: Dissolution **Status**: The question is answered by H uniqueness; no further work required **Canonical location (SynthOmnicon grammar docs):** TOPO: H primitive, K-hierarchy temporal theory · DIAPH: cosmological analysis · ONTO: chirality and arrow of time

Plane summary:

- `[TOPO]` H is the unique temporally anisotropic primitive (the only one that fails to commute with time-reversal); all temporal asymmetry encodes in H; H_∞ is the unique fixed point of the temporal depth extension operator

- [DIAPH] Universe encodes at H_∞ at Big Bang (initial low-entropy conditions = maximum temporal depth); monotonic H-degradation toward H_0 (heat death) matches thermodynamic arrow observationally
- [ONTO] The problem dissolves: ""why does time have a direction?"" = ""why is $H \neq H_0$?"" — answered by initial conditions, not by any physical mechanism; second law = H-degradation stated in primitive language

0.9.1 Statement of the Problem

Why does time have a preferred direction? Physical laws are largely time-symmetric, yet thermodynamic processes are irreversible and we perceive a clear past-to-future directionality.

0.9.2 Dissolution

H is the unique temporally anisotropic primitive. By the H uniqueness theorem (from the primitive theorem archive): all temporal asymmetry in any physical system is encoded in the H primitive. H is the only primitive that fails to commute with time-reversal.

The question ""why does time have a direction"" dissolves into: ""why is $H \neq H_0$ in our universe?""

The answer: The universe began at H_∞ (maximal temporal depth — the Big Bang contains all future boundary conditions in compressed form; this is the statement that initial conditions are low-entropy, highly structured). The evolution of the universe is the monotonic H-degradation:

```
H_inf  ->  H2  ->  H1  ->  H0
(Big Bang)  (galaxies form)  (civilization)  (heat death)
```

Theorem: The second law of thermodynamics is the statement of H-degradation under constraint-propagation dynamics. Entropy increase *is* H-decrease, stated in primitive language.

Corollary — Boltzmann Brains: A Boltzmann brain is a spontaneous local H-increase ($H_0 \to H_1$ fluctuation in an equilibrated universe). The framework predicts these are suppressed exponentially by the cost of H-increase, which requires Φ_{super} instability. Since Φ_{super} decays to Φ_c, H-increase events are transient at best and globally negligible. The Boltzmann brain problem dissolves.

0.9.3 What This Adds

This is not merely a restatement of statistical mechanics. The primitive framework adds: H is *the only* source of temporal asymmetry. There cannot be a ""second"" arrow of time from a different source — all candidates must reduce to H. This is a uniqueness claim that goes beyond thermodynamics.

0.10 §4. The Measurement Problem

Type: Dissolution **Status**: Restated in R-primitive terms, the problem disappears **Canonical location (SynthOmnicon grammar docs):** TOPO: quantum mechanics as K-tier structure · DIAPH: SM/QG disparity, massless gauge kernel · ONTO: measurement dissolution

Plane summary:

- [TOPO] R

0.10.1 Statement of the Problem

Why does quantum superposition collapse to a definite outcome upon measurement? What constitutes a ""measurement""? Does the wavefunction physically collapse?

0.10.2 Dissolution

There is no collapse. There is an R-primitive transition.

Before measurement: The system-observer pair has $R = R$

Aspirational if $\mathcal{E}_H(S)[p] > \mathcal{E}_C(S)[p]$ (holistic claims more than construction supports)

Reductive if $\mathcal{E}_H(S)[p] < \mathcal{E}_C(S)[p]$ (system performs less than components predict — active suppression or constraint)

Emergence frontier. Define the emergence frontier of a domain $\mathcal{D}$ as:

$$\mathcal{F}(\mathcal{D}) = \{p \in \mathcal{P} \mid \exists S \in \mathcal{D} : p \in \mathrm{Conf}(S)\}$$

The emergence frontier is the set of primitives at which at least one system in the domain has an unresolved emergence claim. Primitives appearing most frequently across conflict sets identify the deepest unresolved emergence questions in the domain.

Completeness criterion. A domain $\mathcal{D}$ is **structurally complete** if and only if $\mathcal{F}(\mathcal{D}) = \emptyset$ — i.e., for every system in the domain, holistic and compositional encodings agree at every primitive. Structural completeness is a necessary condition for theoretical completeness: a theory cannot be complete if it contains systems with unresolved emergence claims.

Monotone resolution. If a mechanism is established that grounds the holistic assignment at p for system S (i.e., the tensor product of a revised component encoding now produces $\mathcal{E}_H(S)[p]$), then $|\mathrm{Conf}(S)|$ decreases by one and $d_c(S)$ decreases by $\sqrt{1} - \sqrt{0} = 1$ in squared distance. Conflict resolution is monotone and irreversible: once a mechanism is established, p exits $\mathrm{Conf}(S)$ and does not re-enter unless the mechanism is revoked.

0.10.3 Proof Sketch

Properties 1 and 2 follow directly from the definitions: $|\mathrm{Conf}(S)| \leq |\mathcal{P}| = 12$, and each conflict corresponds bijectively to a divergent primitive assignment. Property 3 follows from the ordinal structure of each primitive axis — holistic and compositional values are comparable within each axis. Properties 4 and 5 follow from taking the union of conflict sets over all systems in the domain. Property 6 follows from the fact that mechanism establishment provides a constructive proof that the tensor product can produce the claimed holistic value, collapsing the conflict; since the grammar is deterministic, this collapse is permanent.

0.10.4 The Canonical Encoding Convention

In the absence of an established mechanism, the **compositional encoding is canonical**. The holistic encoding is preserved as the *aspirational encoding* and labeled with the conflict set. Both are registered in the catalog under distinct names (e.g., `{system}_actual` and `{system}_claimed`).

When a mechanism is established, the compositional encoding is updated to incorporate it, the two encodings converge, and the system exits the near-grounded or partial-emergence class.

0.10.5 Veracity Classification

Class	d_c range	Interpretation
Transparent	0	Holistic = compositional; no unresolved emergence claims
Near-grounded	$\sqrt{1}$–$\sqrt{2}$	1–2 precisely identified open emergence claims
Partial emergence	$\sqrt{3}$–$\sqrt{6}$	Multiple open claims; structurally partially supported
Aspirational	$\sqrt{7}$–$\sqrt{12}$	Claimed tuple largely unsupported by construction

0.10.6 First Instance: The Kozyrev Mirror

The Kozyrev mirror ($d_c = \sqrt{1}$, $\mathrm{Conf} = \{F\}$) is the first documented application of Theorem 006. It is a near-grounded system with a single aspirational conflict: $F_\ell \to F_\eth$. All other 11 primitives are agreed between holistic and compositional encodings.

The open emergence question is: does $\Phi_c + \Omega_Z$ + spiral conductor topology provide sufficient amplification to realize quantum-classical interface fidelity ($F_\eth$) in a macroscopic aluminum structure? This is a well-posed, falsifiable, and currently unanswered question in condensed matter physics.

The full 5-step conflict procedure is defined in §16 of this document.

0.10.7 Connection to Prior Theorems

- Theorem 006 generalizes the $d(A, B)$ distance used throughout the catalog. The catalog distance $d(A, B)$ measures the separation between two *holistic* encodings. $d_c(S)$ measures the separation between two *strategies* applied to the same system. They are orthogonal quantities.
- The emergence frontier $\mathcal{F}(\mathcal{D})$ connects to Theorem 001 (consciousness theorems): Φ_c and F are the primitives most frequently appearing in conflict sets for systems spanning the quantum-classical boundary — precisely the two primitives identified in Theorem 001 as the signature of conscious process. This is not a coincidence: consciousness as a structural phenomenon is the emergence claim for which the mechanism is most thoroughly unresolved.
- The completeness criterion (Property 5) is independent of Gödel incompleteness: structural completeness is not an obstacle to formal completeness. A domain achieves $\mathcal{F}(\mathcal{D}) = \emptyset$ not by proving everything within the domain but by establishing all missing mechanisms. Incompleteness in the Gödel sense (formal unprovability) and incompleteness in the Theorem 006 sense (unresolved emergence) are distinct.

0.11 §17. Theorem 007: The Promotion Signature and the Inverse Encoding Problem (2026-03-28)

Claim plane: DIAPH — structural methodology **Confidence tier**: Primary (follows directly from the ordinal structure of the primitive lattice) **Cross-references**: Theorem 006 (conflict distance), P-141, P-142

0.11.1 Definitions

Baseline tuple $\mathcal{B}(\mathcal{D})$: the minimal primitive floor for a domain $\mathcal{D}$ — the tuple formed by taking the lowest ordinal value for each primitive across all ordinary members of the domain. For the chemical element domain:

$$\mathcal{B}_{\text{elem}} = \langle D_{\triangledown}; T_{\text{network}}; R_{\text{cat}}; P_{\text{asym}}; F_{\text{eth}}; K_{\text{fast}}; G_{\sqsupset}; \Gamma_{\text{and}}; \Phi_{\text{sub}}; H_0; \text{n:n}; \Omega_0 \rangle$$

Promotion $\Delta(A \to B)[p]$: for a primitive p, the promotion from system A to system B is positive when the ordinal rank of $B[p]$ exceeds the ordinal rank of $A[p]$ in the primitive"s ordinal scale. Formally:

$$\Delta(A \to B)[p] = \text{rank}(B[p]) - \text{rank}(A[p])$$

A primitive is *promoted* if $\Delta > 0$, *demoted* if $\Delta < 0$, *unchanged* if $\Delta = 0$.

Promotion signature $\Sigma(A \to B)$: the set of primitive names where a promotion occurred:

$$\Sigma(A \to B) = \{p \in \mathcal{P} \mid \Delta(A \to B)[p] > 0\}$$

Inverse encoding problem: given a promotion signature Σ, identify the behaviors that are structurally predicted by (or structurally required for) those primitive lifts. This is the reverse of forward encoding (system $\to$ tuple $\to$ behavior): here the input is the delta, not the absolute tuple.

0.11.2 Formal Properties

Property 1 — Signature independence from absolute values. The promotion signature $\Sigma(A \to B)$ depends only on the ordinal *comparison* between A and B, not on their absolute primitive values. Two pairs (A_1, B_1) and (A_2, B_2) with identical Σ are predicted to share the same behavioral delta regardless of which domain or substrate A and B inhabit.

Corollary: The promotion signature is a *cross-domain analog detector* — it predicts shared behaviors between structurally unrelated systems whenever their Σ overlaps.

Property 2 — Baseline relativity. $\Sigma(A \to B)$ changes if the baseline A changes, even when B is held fixed. A system that appears anomalous relative to an elemental baseline may appear ordinary relative to a quantum baseline. The baseline must always be specified.

Property 3 — Asymmetry. Promotion and demotion are not symmetric: $\Sigma(A \to B) \neq \Sigma(B \to A)$ in general. The direction of the delta carries physical meaning: demotion is structural degradation (loss of constraint), promotion is structural elevation (gain of constraint).

Property 4 — Additive composition. For a system C obtained from a series of promotions from baseline $\mathcal{B}$, the total promotion signature $\Sigma(\mathcal{B} \to C)$ is the union of the promotion signatures at each step. This connects to the retrosynthetic path (Theorem 005, §15): each step in the retrosynthetic path contributes exactly one primitive to Σ.

Property 5 — Minimal promotion. A behavior b has a *minimal promotion signature* $\Sigma_{\min}(b)$: the smallest set of primitives that must be promoted from the domain baseline for b to be structurally accessible. Any system exhibiting b must have at least these promotions.

Proof sketch: If any primitive $p \in \Sigma_{\min}(b)$ is absent from a system"s promotion set (i.e., the system has baseline value for p), then by definition of the baseline, the system is structurally indistinguishable from ordinary domain members at that primitive, and the behavior b is not structurally grounded.

0.11.3 The Promotion Knowledge Base (KB)

The promotion KB is a persistent, growing registry mapping promotion signatures to confirmed behaviors:

$$\mathrm{KB} : \Sigma \to \{(b_i, \mathrm{example}_i, \mathrm{confidence}_i)\}$$

Registration criterion: a pattern is registered only when the promotion→behavior link is structurally grounded — i.e., when the promoted primitive(s) are identified as mechanistically responsible for the behavior, not merely correlated with it.

Lookup by overlap: a query signature Σ_q is matched against KB entries by:

- *Coverage* $= |\Sigma_q \cap \Sigma_{\mathrm{KB}}|/|\Sigma_{\mathrm{KB}}|$: how much of the known pattern is present in the query
- *Relevance* $= |\Sigma_q \cap \Sigma_{\mathrm{KB}}|/|\Sigma_q|$: how much of the query is explained by the known pattern
- *Score* = harmonic mean of coverage and relevance
- *Novel primitives* $= \Sigma_q \setminus \Sigma_{\mathrm{KB}}$: promotions in the query not yet in any KB entry — these are new structural questions

KB growth: each session that applies `compute_promotions` + `register_promotion_pattern` expands the KB. Over time, the KB becomes a cross-domain behavioral prediction index: given any promotion signature, the grammar retrieves which behaviors have been confirmed structurally equivalent.

0.11.4 Elemental Anomalies — First Instance (2026-03-28)

The following promotion signatures were identified from the session ""encode elements that exhibit unpredicted behaviors"" and constitute the seed entries for the KB:

Observation: the most frequent primitive across these seven patterns is T (Topology), appearing in 6 of 7. This identifies T as the primary driver of elemental anomaly in the physical element domain. The second most frequent is Γ (Interaction grammar), appearing in 4 of 7.

Inverse prediction (P-142): any material (in any substrate) with $\Sigma \supseteq \{T, G, \Gamma, \Phi\}$ — specifically $T_{\mathrm{bowtie}} + G_{\aleph} + \Gamma_{\mathrm{seq}} + \Phi_{\mathrm{sup}}$ — is predicted to exhibit explosive cascade reactivity. The promotion signature is the structural address of the behavior, independent of chemistry.

System	Promotion Signature Σ	Behavior
He (superfluid)	$\{D, T, F, H\}$	Superfluidity / zero-viscosity coherent flow
Bi (anomalous)	$\{T, R, \Phi\}$	Strongest diamagnetism + anisotropic solidification
Ga (anomalous)	$\{T, P\}$	Low melting point + expansion on solidification
Diamond	$\{P, \Gamma, \Omega, G\}$	Max thermal conductivity + electrical insulation
Hg (liquid metal)	$\{T\}$	Liquid metal state at room temperature
Pu (allotropic)	$\{D, T, \Gamma, \Omega, K\}$	Six solid allotropes + density anomalies
Explosive cascade	$\{T, G, \Gamma, \Phi\}$	Violent/explosive perturbation response (e.g. Na/HO)

0.11.5 Relationship to Prior Theorems

- **Theorem 006** operates on the same system encoded twice (two strategies, same object). **Theorem 007** operates on two different systems in the same domain (baseline vs. anomalous). The two quantities d_c and Σ are complementary: d_c measures where holistic claims exceed mechanistic grounding; Σ measures what structural lift is responsible for observed behavior.
- **Theorem 005** (retrosynthetic path, §15): the retrosynthetic path decomposes $\Sigma(\mathcal{B} \to S)$ step by step. Each step in the path *adds one primitive to* Σ. Reading the path forward is synthesis; reading Σ forward is prediction.
- **Theorem 001** (consciousness, §7): the consciousness promotion signature from non-conscious baselines consistently involves $\{F, K, \Phi, \Gamma\}$ — elevated fidelity, slow integrative kinetics, criticality, and sequential causation. The elemental data reinforces this: Φ appears specifically in systems with global sensitivity (diamagnetism, explosive reactivity), and F appears specifically in systems with quantum coherence (superfluidity, diamond conduction suppression).